

NOTE: 5 students in each group, 10 minutes for each group

1. System Modeling Using UML – GROUP 1

- **Introduction to UML:** Explain its purpose and relevance in software engineering.
 - **Types of UML Diagrams:** Focus on:
 1. Structural Diagrams: Class, Object, Component.
 2. Behavioral Diagrams: Sequence, Activity, Use Case.
 - **Steps to Model a Software System:**
 1. Identify system requirements and stakeholders.
 2. Use Use Case Diagrams to define user interactions.
 3. Design Class and Object Diagrams for system structure.
 4. Model workflows using Activity and Sequence Diagrams.
 - **Tools for UML Design:** StarUML, Lucidchart, Visual Paradigm.
 - **Case Study/Example:** Create a UML model for a library management system or online shopping platform.
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2. Simulation in Agile Software Development – GROUP 2

- **Introduction to Agile and Simulation:**
 - How simulation complements Agile practices.
 - **Key Concepts in Agile Simulation:**
 1. Modeling Scrum or Kanban processes.
 2. Simulating story point estimations.
 3. Predicting sprint outcomes and burndown rates.
 - **Steps to Simulate Agile Processes:**
 1. Collect historical data from Agile teams.
 2. Use tools like Jira or Trello to extract parameters.
 3. Simulate scenarios like scope creep or team bottlenecks.
 - **Software Tools:** AnyLogic, Simio.
 - **Real-Life Application:** Simulate a team's sprint cycle and predict timeline deviations.
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3. Queueing Models for Software Systems – GROUP 3

- **Introduction to Queueing Theory:** Overview of queues in software systems (e.g., server requests).
- **Key Concepts:**

1. Types of queues (e.g., FIFO, priority queues).
 2. Parameters: Arrival rates, service times, system capacity.
- **Steps to Build a Queueing Model:**
 1. Define the system (e.g., a web server handling requests).
 2. Identify key metrics: Throughput, latency, drop rate.
 3. Implement a model using tools like MATLAB or Python (SimPy).
 - **Applications:**
 - Load balancing in web servers.
 - Database query handling.
 - **Demonstration:** Simulate a queue for a web server handling concurrent user requests.
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4. Monte Carlo Simulation in Risk Management – GROUP 4

- **Introduction to Monte Carlo Simulation:**
 - Overview and its use in predicting uncertainties.
 - **Steps to Conduct Monte Carlo Simulation:**
 1. Define the problem (e.g., risk of project delays).
 2. Assign probability distributions to uncertain variables.
 3. Run multiple iterations to observe outcomes.
 - **Use Cases in Software Projects:**
 - Cost estimation.
 - Resource allocation.
 - Defect detection probability.
 - **Tools:** Excel (with add-ons), MATLAB, Python (NumPy, SciPy).
 - **Example:** Simulate risks for a mobile app development project.
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5. Discrete Event Simulation in Software Engineering – GROUP 5

- **Overview of Discrete Event Simulation (DES):**
 - Explain events, states, and timelines in DES.
- **Applications in Software:**
 1. Analyzing database transactions.

2. Simulating user behavior in systems.

- **Steps to Build DES Models:**

1. Define events (e.g., request arrival, processing start).
2. Establish state transitions.
3. Model event timelines.

- **Tools:** SimPy, AnyLogic, Arena.
 - **Example:** Simulate transaction handling in an e-commerce checkout system.
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6. Modeling Software Testing Processes – GROUP 6

- **Introduction to Software Testing:**

- Explain static vs. dynamic testing.

- **Why Model Testing Processes?**

- Optimize test case selection and defect detection.

- **Steps to Simulate Testing Processes:**

1. Collect defect rates and testing timelines.
2. Simulate defect discovery over time.
3. Predict test coverage based on resources.

- **Case Study:** Model a testing process for a mobile application.
 - **Tools:** MATLAB, Simulink.
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7. Agent-Based Modeling in Software Systems – GROUP 7

- **Overview of Agent-Based Modeling (ABM):**

- Definition and key components (agents, environment, interactions).

- **Applications:**

- Simulating user behavior in software applications.
- Distributed system optimization.

- **Steps to Build ABM:**

1. Define agent behaviors and rules.
2. Design the environment.
3. Implement and observe emergent behaviors.

- **Tools:** NetLogo, Repast, AnyLogic.
 - **Example:** Simulate traffic flow in a navigation app.
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8. Simulating Software Architectures– GROUP 8

- **Introduction to Software Architectures:**
 - Types (monolithic, microservices, event-driven).
 - **Key Metrics to Analyze:**
 - Latency, scalability, fault tolerance.
 - **Steps to Simulate Architectures:**
 1. Model interactions between components.
 2. Simulate load balancing and failover mechanisms.
 - **Tools:** Docker, Kubernetes (for microservices).
 - **Example:** Simulate a microservices-based e-commerce platform.
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9. Simulation in Cybersecurity – GROUP 9

- **Overview of Cybersecurity Challenges:**
 - Explain common threats (DDoS, phishing, malware).
 - **Steps to Simulate Cybersecurity Scenarios:**
 1. Define attack and defense mechanisms.
 2. Model network traffic patterns.
 3. Simulate and evaluate responses.
 - **Tools:** Kali Linux (penetration testing), Wireshark.
 - **Example:** Simulate a DDoS attack on a server.
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10. Big Data and Predictive Modeling– GROUP 10

- **Introduction to Big Data:**
 - Explain characteristics (volume, variety, velocity).
- **Steps to Build Predictive Models:**
 1. Collect and preprocess data.

2. Build machine learning models (regression, classification).
 3. Simulate predictive scenarios.
- **Tools:** Python (pandas, scikit-learn), Apache Spark.
 - **Example:** Predict user behavior for a streaming service.
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11. Cloud Computing and Resource Optimization – GROUP 11

- **Overview of Cloud Platforms:**
 - Explain IaaS, PaaS, SaaS.
 - **Steps to Simulate Cloud Resource Allocation:**
 1. Define workloads (e.g., web apps, analytics).
 2. Model VM allocation and auto-scaling.
 3. Optimize costs and performance.
 - **Tools:** AWS Cloud Simulator, AnyLogic.
 - **Example:** Optimize resources for a real-time video streaming service.
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12. Software Process Simulation Modeling (SPSM) – GROUP 12

- **Introduction to SPSM:**
 - Explain its role in improving software development processes.
- **Steps to Simulate Software Processes:**
 1. Define processes (e.g., requirements gathering, coding).
 2. Model team dynamics and workflows.
 3. Simulate delays and bottlenecks.
- **Tools:** AnyLogic, MATLAB.
- **Example:** Simulate the process of delivering a mobile app project.